

Homework 2

1. (a.) The derivative of a distribution $f \in \mathcal{S}'$ is defined by the rule

$$\langle \partial f, \varphi \rangle = -\langle f, \partial \varphi \rangle, \quad \text{for } \varphi \in \mathcal{S}.$$

To compute the derivative of $\chi_{[a,b]}$, letting $\varphi \in \mathcal{S}$, we have

$$\langle \partial_x \chi_{[a,b]}, \varphi \rangle = -\langle \chi_{[a,b]}, \partial_x \varphi \rangle = -\int_a^b \partial_x \varphi = \varphi(a) - \varphi(b).$$

So the action of $\partial_x \chi_{[a,b]}$ is the same as $\delta_a - \delta_b$, so this is the distributional derivative.

- (b.) Take $x_j = x$ without loss of generality. Again letting $\varphi \in \mathcal{S}(\mathbb{R}^2)$, we can write

$$\begin{aligned} \langle \partial_x \chi_{B(0,1)}, \varphi \rangle &= -\int_{B(0,1)} \partial_x \varphi(x, y) dV = -\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \partial_x \varphi dx dy \\ &= \int_{\partial B(0,1)^-} \varphi dS - \int_{\partial B(0,1)^+} \varphi dS, \end{aligned}$$

where dS denotes a measure on the boundary $\partial B(0, 1)$ and $-$ and $+$ denote left and right semicircles. This is the difference between the integral of φ over the left and right semicircles.

- (c.) We compute the fourier transforms of $\sin(x)$ and $\cos(x)$, which are not integrable functions in the L^1 sense. We use the definition

$$\langle \mathcal{F} u, \varphi \rangle = \langle u, \mathcal{F} \varphi \rangle, \quad \text{for } \varphi \in \mathcal{S}.$$

We already know that $(e^{2i\pi x \cdot a}) = \delta_a$ for $a \in \mathbb{R}$, so that $(e^{\hat{i}x}) = \delta_{\frac{1}{2\pi}}$. Moreover, $\sin(x)$ can be expressed as

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

so that by linearity of the fourier transform, we have

$$\mathcal{F}(\sin(x)) = \frac{1}{2i} \left(\delta_{\frac{1}{2\pi}} - \delta_{\frac{-1}{2\pi}} \right).$$

- (d.) We need to compute

$$\langle \partial \log |x|, \varphi \rangle = -\langle \log |x|, \partial_x \varphi \rangle$$

for any $\varphi \in \mathcal{S}(\mathbb{R})$. Let $\varepsilon > 0$. Then

$$\int_{\varepsilon \leq |x|} \varphi(x) \frac{1}{x} dx = -\int_{\varepsilon \leq |x|} \partial_x \varphi(x) \log |x| dx$$

after integrating by parts, noting that the first integrand is integrable for any $\varepsilon > 0$. Moreover, the function $\log |x|$ is integrable on $\mathbb{R}$, so we can let epsilon go to zero in the final integral. So the limit converges for all φ :

$$u(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon \leq |x|} \varphi(x) \frac{1}{x} dx = -\int_{\mathbb{R}} \partial \varphi(x) \log |x| dx = \langle \log |x|, \partial_x \varphi \rangle =: \langle \partial_x \log |x|, \varphi \rangle.$$

We have proven the sequence of tempered distributions defining u converges to $\partial_x \log |x|$ in the weak* topology on $\mathcal{S}'(\mathbb{R})$, and so this is the derivative.